

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

27th May 2014

Subject SA2 - Life Insurance

Time allowed: Three hours (14.45* - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which are meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the same should answer the question with reference to Indian environment.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** A well-established Indian life insurance company “A” sells its products through a large Tied Agency force operating from branches located nationwide and has grown rapidly by opening new branches and recruiting and training large number of agents. The company writes it’s with-profit and conventional without profit business in a with-profit fund within which profits are shared between policyholders and shareholders in the ratio 90:10. The company writes unit-linked business in a 0:100 without Profit Fund.

The company under unit linked segment has written only single premium unit-linked investment policies. The policyholders have the options to choose among growth, fixed interest, and balanced funds. The death benefit in the Unit Linked contract is 105% of the unit value. The initial commission is 2% of the single premium and the company also incurs management expenses for procuring and maintenance of business. To meet the expenses the product has a level annual fund management charge of 0.8% per annum and there are no other charges. The average premium income per policy is Rs 100,000.

The company values its policy liabilities using the prospective gross premium valuation method. The company also calculates its embedded value annually, using a traditional Embedded Value approach.

The company carries out the Embedded Value calculations and publishes its Embedded Value results each year before completion of its statutory solvency returns. This year, the Appointing Actuary changed the statutory valuation basis for conventional without profits immediate annuities after the company has published its embedded value results. The actual investment yields have remained unchanged. The Appointed Actuary however, has strengthened the valuation rate of interest and also the mortality assumption in the statutory valuation basis.

- i)** Describe how the changes in following assumptions might affect the embedded value.

a) Changes in valuation rate of interest (4)

b) Changes in the valuation mortality assumption (4)

The company also performs analysis of change on its embedded value. One of the Directors while looking at the results of current year’s analysis of change on its unit linked portfolio has observed that there has been a significant adverse experience variation due to persistency of this line of business. He commented that the assumptions used in calculating embedded values in future should be moved in line with this experience.

- ii)** Outline the issues the company needs to consider before making the suggested change in calculating embedded values in future. (7)

An independent investment analyst commented that the traditional deterministic embedded value calculations that the company uses do not allow appropriately for risk and will soon become obsolete.

- iii)** Discuss the comment and outline the ways to calculate the embedded value that might allow for risk appropriately. (11)

Another Indian life insurance company B which writes both conventional with-profits and without profits business but is closed to new business has projected its solvency position for next three years.

Solvency position – Company B (in crore rupee)	Actual 2013-14	Projected 2014-15	Projected 2015-16	Projected 2016-17
Required Solvency Margin	5,500	5,830	6,180	6,550
Available Solvency Margin	9,960	10,140	10,160	10,180

The company's statutory solvency ratio has declined from over 200% to 181% in less than three years period. Concerned with the projected solvency position of the company and the regulatory provisions in this regard the Board of Company B in its meeting discussed about selling the entire business in its books to another life insurance company.

In order to augment its Life portfolio the Company A is considering making an offer to purchase Company B. The company A would merge all of the without profits business of Company B into its 0:100 Non Profit Fund. A separate with-profit fund within which profits are shared between policyholders and shareholders in the ratio 90:10 would be created in company A account books for the with-profits business of Company B.

- iv)** Describe the statutory procedure for amalgamation of company A with company B. (6)

Company B has approached Company A to discuss the merger, and it has been agreed that the synergies arising from expense savings would be of benefit to both companies. It has now been decided by the Board of company A to purchase company B.

- v)** You are the chief actuary of company A and have been asked by your Chief Executive to estimate the buying Price of Company B. Describe regulatory and other considerations that you would take into account to determine the price to offer for Company B. You can assume that the policy data and other relevant information relating to company B are available to you (10)

Without Profit business of company B constitutes a range of unit-linked funds including growth, fixed interest, and balanced funds.

- vi)** Outline the key issues that Company A needs to consider with respect to calculation of unit prices for the Unit Linked Funds at the merger date post-merger of Company B with Company A. (8)

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Q.2) You are Appointed Actuary of a mid-sized Life Insurance company in India. The Chief Executive of the company believes that the new products Regulations provide an opportunity to create products which could be appealing to the customers. CEO noticed that a significant portion in the regulation was devoted towards Variable Insurance Products. He asks you to develop an individual Non-participating, Non-Linked Variable Insurance plan.

- i)** Describe the regulatory considerations and other factors that you would take into account while designing a Non Linked Non-Participating, Variable Insurance Product (12)

The Chief Executive of the company observes that there is some downside protection from the underlying guarantee in a VIP (Variable Insurance) product and therefore the company should have some exposure in equity to improve the expected return to the VIP policyholder.

- ii)** Outline the regulatory and other non-regulatory considerations and restrictions that you will need to take into account for investment management including the choice of assets for proposed VIP fund. (9)

- iii)** Describe the considerations that you will take into account, giving reference of respective regulations, while valuing the Liabilities in respect of proposed VIP Product. (8)

As part of internal product approval process, you are required to capture the risks and their management arising from a new product proposition and possible mitigation options

- iv)** Describe the risks emerging from the proposed VIP product including ALM risks. Outline how you would propose to manage or mitigate these risks. (12)

As a product documentation process you are required to capture the details of the regulatory reporting with respect to the product.

- v)** Summarise the reporting requirements applicable to VIP Non-Par non linked product. (5)

- vi)** Describe how you would calculate the solvency requirement in respect of proposed VIP Product specifying relevant provisions of insurance act and IRDA regulations and circulars. (4)

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